

Lesson 8 Assessment

Integrating Multiple APIs Together — n8n

1. Assessment Overview

This assessment evaluates the ability to design, build, test, and document a complete n8n API orchestration workflow by integrating multiple APIs in a single automation.

2. Assessment Objective

Demonstrate an end-to-end API orchestration workflow that initializes a support ticket, generates a transaction ID, performs customer lookup, logs ticket data to a mock database API, sends a notification using ancestor variables, and validates the complete execution flow.

3. Implemented n8n Workflow

Step	Node / Function	Purpose
1	Start (Manual Trigger)	Starts the workflow manually.
2	Initialize Ticket Payload	Creates the initial support ticket data.
3	Generate Transaction ID	Creates a unique transaction identifier dynamically.
4	Customer Lookup	Maps customer information, including email to customer ID.
5	Log Ticket to Mock DB API	Sends ticket details to the mock database API.
6	Verify DB Response	Checks the API response from the database logging step.
7	Send Notification	Sends a confirmation request using workflow/ancestor variables.
8	Validate Execution Flow	Validates the final workflow output and successful execution.

4. Practical Assessment Checklist

- Workflow created successfully
- Support ticket payload initialized
- Transaction ID generated dynamically
- Customer lookup configured
- Database API integration completed
- Notification API configured
- Workflow executes successfully end-to-end
- Execution verified
- Source workflow prepared for export
- Demo video prepared for submission

5. Validation Result

The provided n8n workflow screenshot shows the complete multi-step API orchestration connected from the Manual Trigger through ticket initialization, transaction generation, customer lookup, mock database logging, response verification, notification, and final execution validation. The execution panel shows a successful run.

6. Success Criteria

- Build a complete multi-step API orchestration workflow.
- Integrate multiple APIs in one automation.
- Use dynamic expressions and ancestor node variables.
- Validate successful execution from start to finish.
- Prepare the required source-code and demonstration deliverables.

7. Submission Deliverables

Deliverable	Status
Loom / YouTube Demo Video	Required
Exported n8n Workflow / Source Code	Required
GitHub Repository	Recommended
Description / Additional Comments	Recommended
Supporting Documents	Optional

8. Implementation Notes

- **Challenges:** Coordinating data between multiple API steps and ensuring values from earlier nodes were correctly referenced.
- **Assumptions:** The mock API endpoints are available and return the expected response structure during execution.
- **Limitations:** The workflow uses mock/API endpoints for assessment purposes rather than a production customer database or notification provider.
- **Enhancements:** Production implementation could add stronger validation, authentication, retry handling, structured logging, and monitoring.

9. Final Submission Checklist

- Completed all assessment exercises
- Verified successful workflow execution
- Prepared workflow/source code
- Prepared Loom/YouTube demonstration
- Prepared assessor comments
- Reviewed deliverables before LMS submission

Assessment: Lesson 8 — Integrating Multiple APIs Together

Platform: n8n

Workflow type: Multi-API orchestration